**27** A nichrome wire, P, has a length of 1.0 m and a diameter of 1.0 mm.

A second nichrome wire, Q, has a length of 2.0 m and a diameter of 2.0 mm.

Which statement correctly compares the resistance of Q with that of P?

**A** The resistance of Q is one-quarter the resistance of P.

**B** The resistance of Q is one-half the resistance of P.

**C** The resistance of Q is the same as the resistance of P.

**D** The resistance of Q is four times the resistance of P.